



Sec: Sr. IIT

Race -14

Date: 29-02-24

MATHEMATICS**Max. Marks : 110****SECTION – I****(MULTIPLE CORRECT ANSWER TYPE)**

This section contains 5 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +5 for correct answer, 0 if not attempted and -2 in all other cases.

- Let $f(x): R^+ \rightarrow R^+$, $g(x) = f(x) \cdot f\left(\frac{2}{x}\right)$, $f(x) \cdot f\left(\frac{2}{x}\right) = x$, $f(\sqrt{2}) = 1$ then

A) $g(x)$ is constant function B) $f(x)$ is constant function

C) $f(x)$ is monomial D) $f'(1) = 1 + g'(1)$
- Let the equation of a conic is $(x-2)^2 + (y-2)^2 = (x+y)^2$

A) The centre of the conic is $(-2, -2)$

B) The eccentricity of the conic is more than '1'

C) The radius of the auxiliary circle of the conic is '4'

D) The angle between the tangent lines drawn from $(0,0)$ to the conic is $\frac{\pi}{2}$
- Which of the following are (is) TRUE

A: $\{a_1, a_2, \dots, a_{10}\}$, $B: \{b_1, b_2, \dots, b_5\}$ number of on- to mappings of $f: A \rightarrow B$ such that $f(a_1) \leq f(a_2) \leq \dots \leq f(a_{10})$ is 126

B) The sum of the squares of the roots of $x^6 - x^5 + x^4 - x^3 + x^2 - x + 1 = 0$ is "-1"

C) Let $\alpha, \beta, \gamma > 0$, $\alpha + \beta + \gamma = \sqrt{3}$ then the minimum value of $\frac{\pi}{\tan^{-1}\alpha + \tan^{-1}\beta + \tan^{-1}\gamma}$ is '2'

D) Domain of $\sqrt{x^{10} - x^7 + x^4 - x + 1}$ is set of all Real
- Let DABC is a tetra hedron $\overline{AD}$ is perpendicular to the base ABC, $\angle ABC = \frac{\pi}{6}$ the volume of the tetrahedron is 18, given that $AB + BC + AD$ is minimum possible then

A) The height of the tetra hedron w.r.t the base ABC is 12

B) The length of AC = $3\sqrt{2}(\sqrt{3} - 1)$

C) The height of the tetrahedron w.r.t the base ABC is 6

D) The length of AD = $3\sqrt{2}(\sqrt{3}-1)$

5. Let $\alpha = \tan^{-1}1 + \frac{1}{2}\tan^{-1}2 + \frac{1}{3}\tan^{-1}3$, $\beta = \tan^{-1}1 + 2\tan^{-1}\frac{1}{2} + 3\tan^{-1}\frac{1}{3}$, $|\alpha - \beta| = P\frac{\pi}{Q} + \frac{r}{s}\cot^{-1}3$

($P, Q, r, s \in N$) (in their lowest form) then

A) $P+Q=29$

B) $Q+r=29$

C) $r+s=11$

D) $s+p=11$

SECTION - II

(MATRIX MATCHING ANSWER TYPE)

This section contains 5 questions. Each question has four statements (A, B, C and D) given in Column I and four statements (P, Q, R and S) in Column II. Any given statement in Column I can have correct matching with ONE or MORE statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in Q and R, then for the particular question darken the bubbles corresponding to Q and R in the OMR sheet. For each correct matching **will be awarded +3 marks ONLY and 0 if not attempted and -1 in all other cases.**

6. Match the following

LIST-I		LIST-II	
A)	$g(x) = \left(4\cos^4 x - 2\cos 2x - \frac{\cos 4x}{2} - x^7\right)^{\frac{1}{7}}$ Then $g(g(1)) = \dots\dots\dots$	P)	1
B)	Let $y = x \cdot \frac{dy}{dx} + \frac{dx}{dy}$ then $\left(\frac{dy}{dx}\right)^2$ at $\left(x = \frac{1}{2}\right)$ is	Q)	2
C)	let $x + y = z$, then the value of $\cos^2 x + \cos^2 y + \cos^2 z - 2\cos x \cdot \cos y \cdot \cos z$	R)	3
D)	The curve $\left(\frac{x}{a}\right)^n + \left(\frac{y}{b}\right)^n = 2$ touches the line $\frac{x}{a} + \frac{y}{b} = 1$ At (a,b) then the possible value of "n" = _____	S)	4

7. Match the following

LIST-I		LIST-II	
A)	$\sum_{r=1}^{10} \frac{1}{r} ({}^{20}C_{2r-1}) =$	P)	0
B)	Number of solutions of $e^{-x} = \frac{4}{x}$	Q)	2
C)	$\forall P \in R, (p^2 + 2)x^2 - 2(a+p)x + p^2 + 2p + b = 0$ has Root '1' ($a, b \in R$) then $ a-b $	R)	3
D)	$f(x) = \left(\left((x-2)^2 - 2\right)^2 - 2\right)^2 \dots\dots (2018 \text{ times})$ then $f(0) =$	S)	4

8. Match the following

LIST-I		LIST-II	
A)	$\sqrt{\sin x} : \left[0, \frac{\pi}{3}\right] \rightarrow [0, 1]$	P)	one -one
B)	$\frac{x+3}{x-1} : (1, \infty) \rightarrow (1, \infty)$	Q)	many one
C)	$\sin x : \left[-\frac{\pi}{2}, \frac{4\pi}{3}\right] \rightarrow [-1, 1]$	R)	in to
D)	$\frac{x^2}{x-2} : (2, \infty) \rightarrow (8, \infty)$	S)	on to

9. Match the following

LIST-I		LIST-II	
A)	$y = a_1x + b, y = a_2x + b$ ($a_1 \neq a_2$) meet the coordinate axes at concyclic point then $a_1 a_2 =$	P)	1
B)	If the chord of contact of tangent to $x^2 + y^2 = 4$ from any point of $x^2 + y^2 = a_1^2$ touches the circle $x^2 + y^2 = a_2^2$ then $a_1 a_2 =$	Q)	2
C)	let a_1, a_2 are the shortest largest distances from the point (8,5) To the circle $x^2 + y^2 - 4x + 6y - 12 = 0$ then $\frac{a_2}{a_1} =$	R)	8
D)	Number of integral values of "k" so that the graphs $y = 2k + 1, y = (k - 6)x^2 - 2$ Do not intersect	S)	4

10. Match the following

	LIST-I		LIST-II
A)	$\sin \frac{\pi}{21} \cdot \sin \frac{2\pi}{21} \cdot \sin \frac{3\pi}{21} \dots \sin \frac{10\pi}{21}$	P)	$\frac{\sqrt{11}}{2^{10}}$
B)	$\sin \frac{\pi}{22} \cdot \sin \frac{2\pi}{22} \cdot \sin \frac{3\pi}{22} \dots \sin \frac{10\pi}{22}$	Q)	$\frac{1}{2^{10}}$
C)	$\cos \frac{\pi}{21} \cdot \cos \frac{2\pi}{21} \cdot \cos \frac{3\pi}{21} \dots \cos \frac{10\pi}{21}$	R)	$\frac{\sqrt{11}}{2^{10}}$
D)	$\cos \frac{\pi}{22} \cdot \cos \frac{2\pi}{22} \cdot \cos \frac{3\pi}{22} \dots \cos \frac{10\pi}{22}$	S)	$\frac{\sqrt{21}}{2^{10}}$

SECTION – III

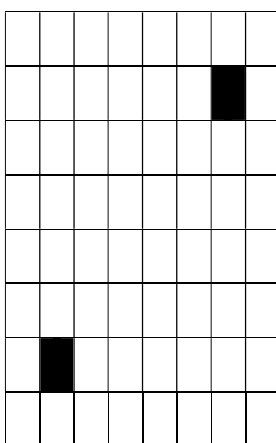
(INTEGER ANSWER TYPE)

This section contains 5 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).

Marking scheme +5 for correct answer , 0 if not attempted and -3 in all other cases.

11. $\lim_{x \rightarrow 0^+} \frac{\left((x^2)^x - 1\right)^2}{e^{(x^x-1)} - (x^x)} = \text{-----}$

12. Consider normal chess board the number of rectangles which includes neither of two shaded small squares is “k” then the values of $\frac{k}{230} =$



13. Let $s = \sum_{k=1}^{89} \cos^6(k)$ then the value of $\left\lceil \frac{s}{10} \right\rceil$ (here $\lceil \cdot \rceil$ denotes GIF) = -----

14. Let $f(x) > 0, \forall x > 0, f(x)$ is derivable and $g(x) = \lim_{n \rightarrow \infty} \left(\frac{f\left(x + \frac{1}{n}\right)}{f(x)} \right)^n$. Then the value of

$$\frac{d}{dx} \left(\int \log_e^{g(x)} dx - \log_e^{f(x)} \right) = \text{-----}$$

15. Let $f(x) = (\log_e x)^2 - \int_1^e \frac{f(t)}{t} dt$ then the area of the region bounded by $y = f(x)$, the tangent line at $x = e$ w.r.t $y = f(x)$ and $x = 1$ is "A" then the value of [A] (here [.] denotes GIF) = -----

PHYSICS

Max. Marks : 110

SECTION – I

(MULTIPLE CORRECT ANSWER TYPE)

This section contains 5 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

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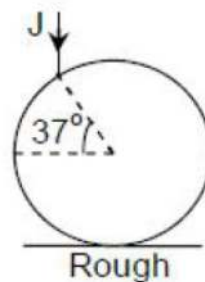
16. A sharp vertical impulse J is given to a uniform solid sphere of mass m and radius R placed on a rough horizontal surface. Choose the correct options.

A) Impulse imparted to the sphere by the surface perpendicular to it will also have magnitude J

B) Minimum value of coefficient of friction between sphere and surface to avoid slipping is $\mu = \frac{4}{7}$

C) If $\mu = \frac{11}{20}$, sphere rolls with backward slipping

D) If $\mu = \frac{13}{20}$, sphere rolls with forward slipping



17. A uniformly charged ring having total charge Q and of radius R is fixed in such a way that plane of the ring is horizontal. A charged particle of charge q and mass m is in equilibrium at any point lying on y -axis under the combined forces of electric field and earth's gravitational field. Choose the correct statements.

A) If $qQ > 0$, particle must be on $+y$ -axis

B) If $qQ < 0$, particle must be $-y$ -axis

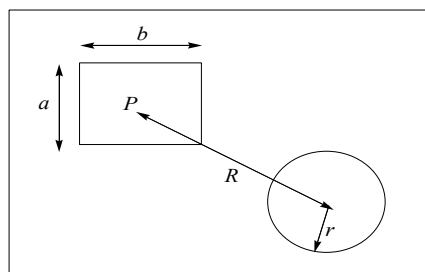
C) If the particle is in stable equilibrium magnitude of y-coordinate of particle must be greater than

$$\frac{R}{\sqrt{2}}$$

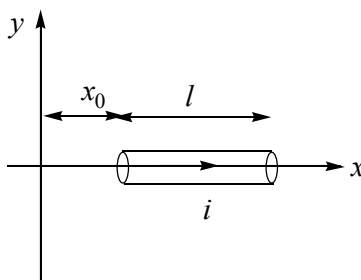
D) If the particle is at rest at a distance y from centre of the ring. Specific charge of the particle is

$$\frac{4\pi\epsilon_0 (y^2 + x^2)^{3/2} g}{Qy}$$

18. There is a rectangular metal plate in which two cavities in the shape of rectangle and circle are made with dimensions as shown. P, Q are the centres of cavities. On heating the plate which of the following quantities increase?



- A) πr^2 B) ab C) R D) b
19. Two blocks, A and B, of same masses, are resting in equilibrium on an inclined plane having inclination with horizontal, $= \alpha (> 0)$. The blocks are touching and exerting non zero normal force on each other with block B higher than A. Coefficient of static friction of A with incline = 1.2 and that of B = 0.8. If motions not imminent
- A) $\alpha < 30^\circ$ B) (Friction)_A > (Friction)_B
- C) $\alpha < 45^\circ$ D) (Friction)_A = (Friction)_B
20. A wire of length l , carrying a current i is kept along x-axis as shown in the figure.



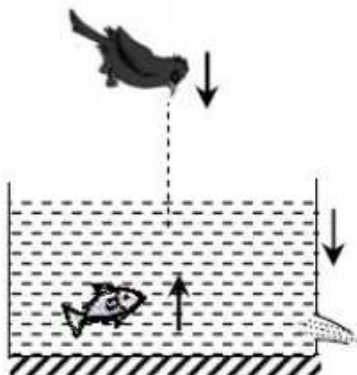
- A) Unit vector along magnetic field due to the wire at (a, b, c) is independent on x_0 .
- B) Unit vector along magnetic field due to the wire at (a, b, 0) is dependent on x_0 .
- C) Unit vector along magnetic field due to the wire at (0, b, c) is independent of b and c.
- D) Unit vector along magnetic field due to the wire at (a, 0, c) is independent of a and c.

SECTION - II

(MATRIX MATCHING ANSWER TYPE)

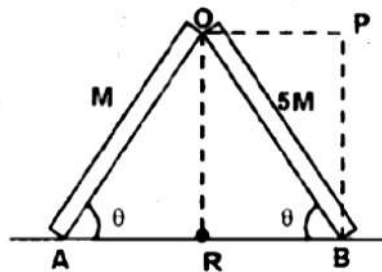
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21. A bird in air is diving vertically over a tank with speed 6cm/s . Base of tank is silvered. A fish in the tank is rising upward along the same line with speed 4cm/s . Water level is falling at the rate of 2cm/s . (Refractive index of water = $4/3$). Then match the entities given in the Column-I with those in Column-II.



COLUMN – I	COLUMN – II
A) The speed of the image of the fish as seen by the bird directly	P) 12
B) The speed of the image of fish formed after the reflection from the mirror as seen by the bird	Q) 4
C) The speed of the image of the bird relative to the fish looking upwards	R) 9
D) The speed of the image of the bird relative to the fish looking downward in the mirror	S) 3

22. There are two rods OA and OB, each of same length l but different masses M and $5M$ respectively, supported against each other. There is a sufficient friction f_1 and f_2 between the ground and feet of left and right rods respectively whereas no friction can act between point contact of the rods. The normal contact force between the ground and feet of left and right rods are N_1 and N_2 whereas normal contact force between top points of the rod is N . Now match the following.

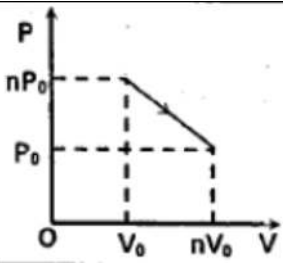
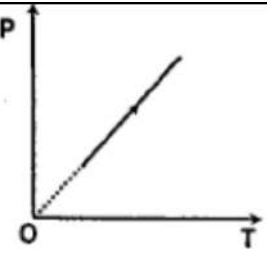
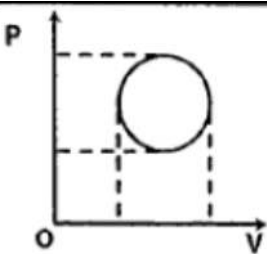
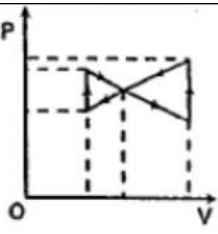


COLUMN – I	COLUMN – II
A) Torque of N acting on OB about P	P) $Mgl \cos \theta$
B) Torque of N acting on OA about R	Q) $\frac{3}{2} Mgl \cos \theta$
C) Torque of friction acting at B about P	R) $\frac{1}{2} Mgl \cos \theta$
D) Torque of weight of OA about A	S) Clockwise

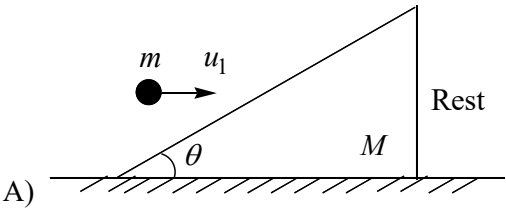
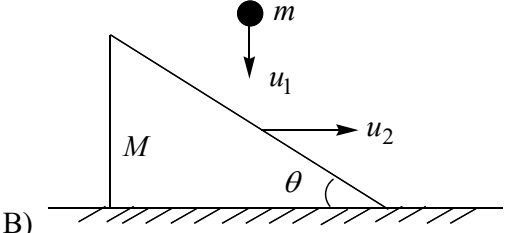
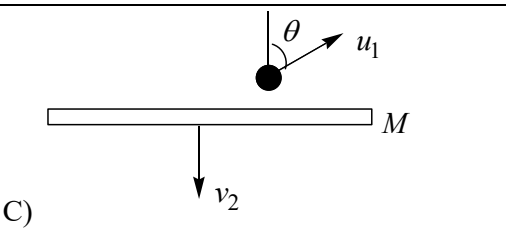
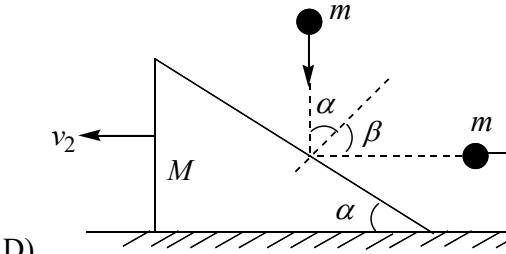
23. In the situations shown in the figure surfaces are frictionless. Match the maximum extension of the springs in each entity in Column-I with those in Column-II, if blocks are initially at rest and springs are initially in natural lengths.

COLUMN – I	COLUMN – II
A)	P) $\frac{8F}{3k}$
B)	Q) $\frac{14F}{3k}$
C)	R) $\frac{2F}{k}$
D)	S) $\frac{4F}{k}$

24. Match the following:

COLUMN – I	COLUMN – II
A) 	P) $W > 0$
B) 	Q) $\Delta Q > 0$
C) 	R) $\Delta U = 0$
D) 	S) $\Delta U > 0$

25. Consider the following situations (All surfaces of contact are smooth).

COLUMN – I	COLUMN – II
A) 	P) Velocity of separation is $v_1 \cos \theta + v_2$
B) 	Q) Velocity of approach is $u_1 \sin \theta$
C) 	R) Velocity of separation is $v_1 \cos \beta + v_2 \sin \alpha$
D) 	S) Velocity of approach is $u_1 \cos \theta + v_2 \sin \theta$

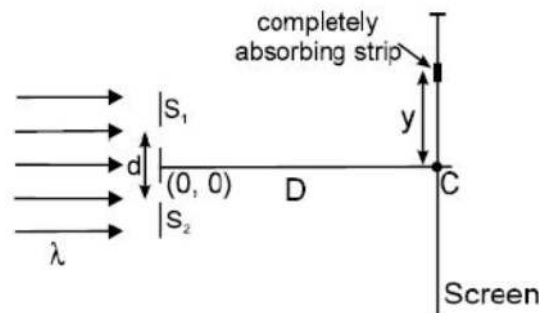
SECTION – III

(INTEGER ANSWER TYPE)

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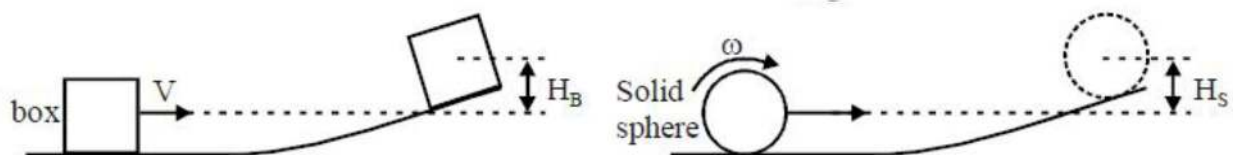
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26. Figure shows two, identical narrow slits S_1 and S_2 . A very small completely absorbing strip is placed at distance y from the point C. C is the point on the screen equidistant from S_1 and S_2 . Assume $\lambda \ll d \ll D$, d and D have usual meaning. When S_2 is covered the force due to light acting on strip is f and when both slits are opened the force acting on strip is $2f$. Find the minimum positive y ($\ll D$) coordinate (in

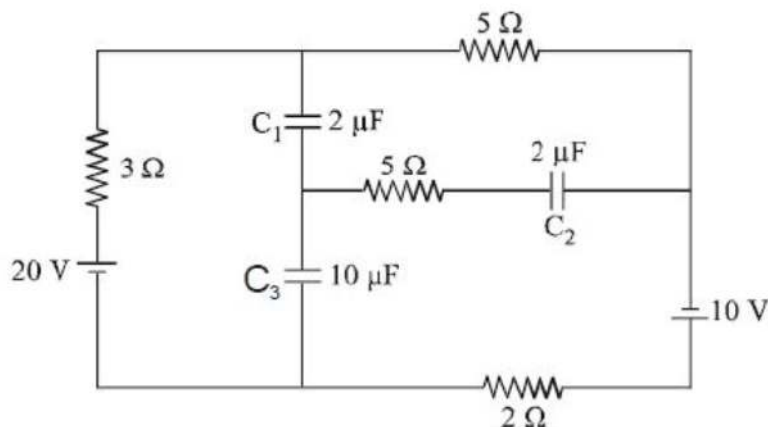


cm) of the strip if $\frac{\lambda D}{d} = 4\text{cm}$.

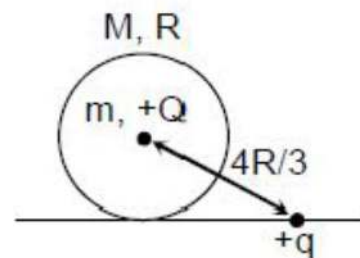
27. A box and a solid sphere of equal mass are moving with the same velocity across a horizontal floor. The sphere rolls without slipping and the box slides without friction. They encounter an upward slope in the floor and each move up the slope some distance before momentarily stopping and then moving down again. The maximum vertical heights reached by the box and the sphere are H_B and H_S respectively. What is the ratio $\frac{5H_S}{H_B}$?



28. In the circuit shown in figure, if the charge on capacitor C_2 in steady state is $\frac{10x}{3} \mu C$, then find value of x .



29. There is a disc of mass M and radius R . A point charge $+Q$ and of mass m is fixed at the centre of the disc. The disc is held on a fixed horizontal rough surface. Another point charged $+q$ is fixed on the surface such that distance between $+q$ and $+Q$ is $\frac{4R}{3}$. Now the disc is set free. If the disc does not loose contact with the ground at the same instant the relation must hold is $M > \eta m$. Find η . (Use $\frac{27Qq}{256\pi\epsilon_0 R^2} = 3mg$)



30. A hydrogen atom moving at a speed v_0 a photon of wavelength 6nm and stops. Mass of hydrogen atom is $1.67 \times 10^{-27} \text{kg}$. When calculated v_0 comes out to be $\frac{13}{n} \text{m/s}$. Find n nearly.

SECTION – I

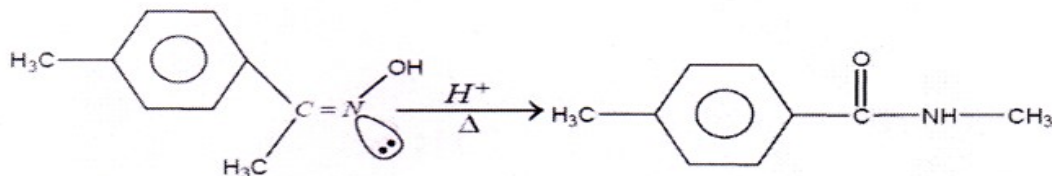
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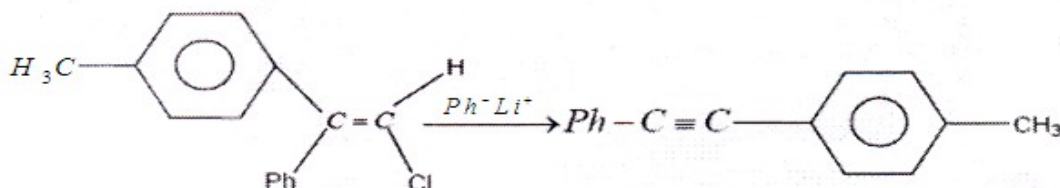
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31. Which of the following reactions represent major products?

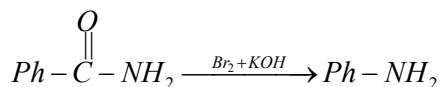
A)



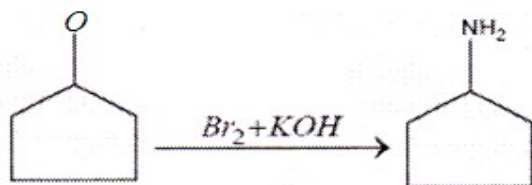
B)



C)



D)



32. Which of the following react with Grignard reagent to form secondary alcohol?

- A) Acetaldehyde B) Formaldehyde C) Methyl formate D) Methyl acetate

33. Which of the following give no residue on heating?

- A) NH_4NO_3 B) NH_4NO_2 C) $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$ D) NH_4Cl

34. $\text{Ag}_{(s)} / \text{AgBr}_{(s)} / \text{Br}^- (0.01\text{M}) \parallel \text{Cl}^- (0.2\text{M}) / \text{AgCl}_{(s)} / \text{Ag}_{(s)}$ if K_{sp} of $\text{AgBr}_{(s)}$ and $\text{AgCl}_{(s)}$ at 25°C are 10^{-12} & 10^{-10} respectively find which is/are true for above cell at 25°C ($\log 2=0.3$)

A) As $E^\circ_{Ag^+/Ag}$ is not given so E_{cell} can not be determined

B) E_{cell} is 0.0413 volt

C) $\Delta G < 0$

D) $E^\circ = 0$

35 At critical temperature

A) $d_{liquid} = d_{vapour}$

B) $d_{liquid} > d_{vapour}$

C) boundary between liquid and vapour phase disappears

D) a transition state exists between liquid and vapour state

SECTION - II

(MATRIX MATCHING ANSWER TYPE)

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36. Match the reactions and reagents give in column-I with the metal ions given in column-II

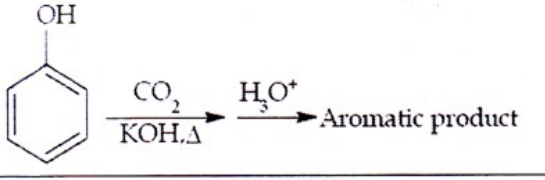
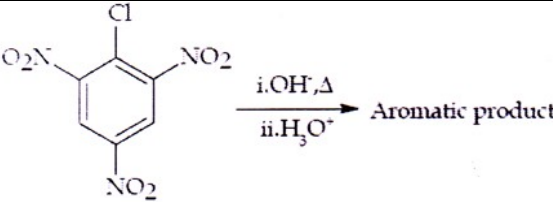
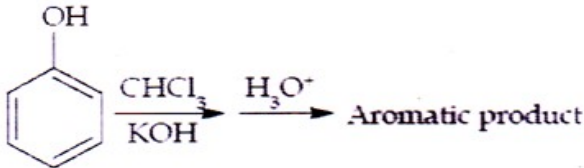
Column I		Column II	
(A)	$Cation(NH_4)_2S \rightarrow \text{metal Sulphide} \downarrow$ Black $\xrightarrow[\text{HNO}_3]{\text{Hot and dil}}$ Soluble	P)	Pb^{2+}
(B)	$Cation + Na_2S_2O_3 \rightarrow \text{Metal Thiosulphate} \downarrow$ q) Cu^{2+} White $\xrightarrow{\text{Excess of hypo}} \text{Soluble}$	(q)	Cu^{2+}
(C)	$Cation + KCN \rightarrow \text{Metal cyanide} \downarrow$ White $\xrightarrow{\text{Excess KCN}} \text{Soluble}$	(r)	Hg^{2+}

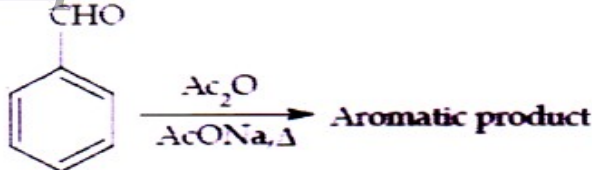
(D)	$\text{Cation} + \text{KI} \rightarrow \text{Metal iodide} \downarrow$ <i>Coloured</i> $\xrightarrow{\text{Excess KI}} \text{Soluble}$	(s)	Ag^+
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37. Match the following Column, Column I with Column II

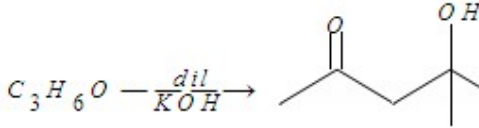
Column I		Column II	
(A)	$[\text{Ma}_3\text{b}_2\text{c}]$	(p)	All stereoisomers are optically inactive
(B)	$[\text{Ma}_3\text{b}_3]$	(q)	Number of geometrical isomers = 2
(C)	$[\text{Ma}_3\text{bcd}]$	(r)	Number of geometrical isomers = 4
(D)	$[\text{Ma}_4\text{bc}]$	(s)	Total 3 Stereoisomers

38. Matrix matching

Column I		Column II	
(A)		(p)	Acid base reaction is an intermediate
(B)		(q)	The aromatic product liberate CO_2 from aq. NaHCO_3 .
(C)		(r)	The aromatic product shows diastereoisomerism

(D)		(s)	All atoms can lie on the same plane in the aromatic product
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39.

Column-I	Column-II
A) $A \xrightarrow[\text{NaOH}]{\text{Conc}} \text{PhCH}_2\text{OH} + \text{Ph}-\text{COO}^- \text{Na}^+$	P) A gives positive Tollen's test
B) 	Q) A gives positive Fehlings test
C) $A \xrightarrow[\text{ii) } \text{H}_3\text{O}^+]{\text{i) } \text{CH}_3\text{COO}^- \text{Na}^+ / \Delta} \text{Ph}-\text{CH}=\text{CH}-\overset{\text{O}}{\parallel}{\text{C}}-$	R) A gives positive iodoform test
D) $A \xrightarrow[\text{H}^+ \Delta]{\text{CH}_3-\text{COOH}} \text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{O}-\text{CH}_2-\text{CH}_3$	S) A gives positive 2-4-DNP test

40. Match the following

Column-I	Column-II
A) Reversible adiabatic compression	P) $\Delta S_{\text{system}} > 0$
B) Reversible vaporization	Q) $\Delta S_{\text{system}} < 0$
C) Adiabatic free expansion of ideal vacuum	R) $\Delta S_{\text{surrounding}} < 0$
D) Dissociation of $\text{CaCO}_3(s) \rightarrow \text{CaO}(s) + \text{CO}_2(g)$	S) $\Delta S_{\text{surrounding}} = 0$

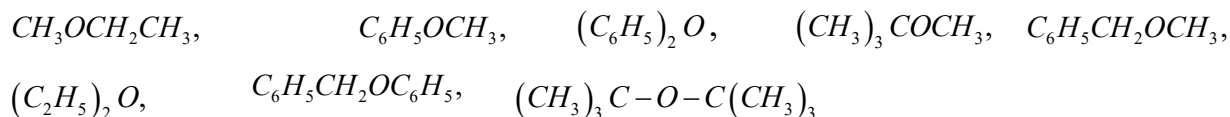
SECTION – III

(INTEGER ANSWER TYPE)

This section contains 5 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).

Marking scheme +5 for correct answer , 0 if not attempted and -3 in all other cases.

41. How many of the following ethers cannot be prepared by williamson's synthesis?



42. How many of the following statements are correct

(A) Nascent hydrogen is a powerful reducing agent than molecular hydrogen.

(B) White phosphorous is more reactive than red phosphorous.

(C) The reactivity of K_2O towards water is more than that of Na_2O towards water

(D) Nitrate salt of silver is soluble in ammonia.

(E) Liquid nitrogen does not stick to the poles of magnet whereas liquid oxygen does

(F) NO_2 is brown in colour at room temperature but colourless below 0°C

(G) Bond order of CN and N_2 are equal but CN is more reactive

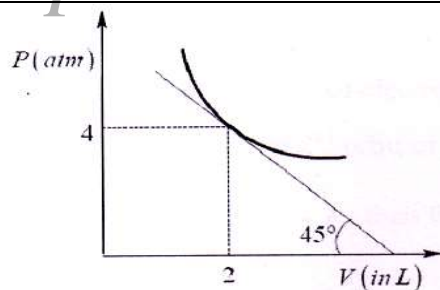
(H) NO^+ is having lower bond strength than NO

43. How many ores are sulphide ore in the following?

i) Azurite ii) Chalcocite iii) Iron pyrites iv) Limonite

v) Malachite vi) siderite vii) pyrolusite viii) Argentiteix) Cinnabar

44. The molar heat capacity of the ideal monoatomic gas undergoing the process shown in the figure is 'x'. The value of '2x' will be



45. If De-Broglie wavelength of electron revolving in 6th orbit of Li^{2+} ion is λ_1 , that of electron revolving in 2nd orbit of He^+ ion is λ_2 , and that for electron revolving in 3rd orbit of 'H' atom is λ_3 , then the value of $\frac{\lambda_3 - \lambda_2}{\lambda_1}$ is

KEY SHEET

MATHEMATICS

1	ACD	2	ABC	3	ABCD	4	BC	5	ABCD
6	A-P B-S C-P D-PQRS	7	A-Q B-P C-Q D-S	8	A-PR B-PS C-QS D-QS	9	A - P B - S C - Q D - Q	10	A-S B-R C-P D-R
11	8	12	4	13	2	14	0	15	0

PHYSICS

16	ABC	17	ABCD	18	ABD	19	BC	20	AD
21	A-R; B-S; C-P; D-Q	22	A-PS; B-Q; C-QS; D-RS	23	A-S; B-S; C-Q; D-R	24	A-PQR; B-QS; C-PQR; D-R	25	A-Q; B-S; C-P; D-R
26	1	27	7	28	3	29	2	30	2

CHEMISTRY

31	ABC	32	AC	33	ABD	34	BCD	35	ACD
36	A-PQS B-PQRS C-QRS D-PR	37	A-PS B-PQ C-R D-PQ	38	A-PQS B-PQS C-PS D-PQRS	39	A -PS B - RS C - PS D - R	40	A-S B-PR C-PS D-PR
41	2	42	6	43	4	44	7	45	1